

Stack Overflow requires external JavaScript from another domain, which is blocked or failed to load. Retry using another source.

What is meant by a "lightweight vm" as discussed in the technology stack for WSL2?

Asked 6 years, 9 months ago Modified 6 years, 9 months ago Viewed 2k times



3



My understanding is that Docker on Windows currently uses a "regular VM" under the hood. WSL2 ([and Docker](#)) will switch to using a lightweight VM. But what does this actually mean; is it just using a smaller initial memory foot print with some memory passthrough technique, or is there more to it?

virtualization

hyper-v

windows-subsystem-for-linux

Share Improve this question Follow

asked Jun 19, 2019 at 19:50



bbarker

13.5k

11

49

70

1 Answer

Sorted by: Highest score (default)



TL;DR

1



The big change is in the move from a **virtualized Linux system call interpreter for the Windows kernel in WSL** to a **full-on Linux kernel provided in WSL2**. This move dramatically cuts down on virtualization overhead.

Juicy Details



Directly from the [DevBlogs Post on the announcement of WSL2](#):

Microsoft will be shipping a Linux kernel with Windows ... This kernel has been specially tuned for **WSL 2**. It has been **optimized for size and performance to give an amazing Linux experience on Windows**.

This is a departure from the ways of the current (*as of writing*) WSL which doesn't make use of a **proper Linux kernel**, demonstrated in the [original WSL overview from 2016](#).

Stack Overflow requires external JavaScript from another domain, which is blocked or failed to load. Retry using another source.

The WSL LXC service runs an interpreter of sorts for native Linux system calls as well as running its own VolFs and DriveFs operations to provide file access between WSL and Windows 10, which essentially performs the role of a traditional VM's translation layer the likes of VirtualBox.

 [WSL components chart](#)

Citation: [MSDN Blog](#)

Little is known as of yet about the exact system employed for WSL2, what we do know is from the [Build2019 WSL2 talk](#). To help answer the question regarding file system changes and the light VM:

 [Timestamp: 29:58](#)

Here, we see that the Linux kernel runs alongside the NT Kernel instead of as a virtualized environment on top of it. (*as a Windows service*). The lightweight VM likely comes into play for facilitating the necessary interactions between the two kernels.

 [Timestamp: 41:11](#)

This gives a peek into the inner workings of that interoperability layer. Discussed verbally in the Build2019 talk, the two kernels serve each other files via natively hosted file servers (*inaccessible to the Windows userspace via means other than WSL2*).

Again, much is still up in the air from our perspective as users due to the limited details currently available to us at the time of writing.

Share Improve this answer Follow

answered Jun 20, 2019 at 18:56

 ifconfig
ifconf 6,998 10 50 69

! Sign up to request clarification or add additional context in comments.



3 Comments ▼

Add a comment



bbarker Over a year ago

Sounds like the memory at least is how I imagined it ("only use what is needed").

▲ 0 Reply ...



bbarker Over a year ago

Stack Overflow requires external JavaScript from another domain, which is blocked or failed to load. Retry using another source.

▲ 0 [Reply](#) [...](#)



[ifconfig](#) Over a year ago [✎](#)

@bbarker Sounds like you know a lot more about the hypervisor layer than I do! :) BTW: If the answer is to your liking, labeling it as accepted would be great!

▲ 0 [Reply](#) [...](#)

Start asking to get answers

Find the answer to your question by asking.

[Ask question](#)

Explore related questions

[virtualization](#) [hyper-v](#) [windows-subsystem-for-linux](#)

See similar questions with these tags.